

FL-02: Prompt Engineering Iteration Log & Cross-Model Audit

Author: Rida Eman (CS Senior & ML Intern) | Track: AI Fluency (Foundations) | Repo: github.com/ridaeman02/flyrank-ml-internship

1. Target Task Context (From FL-01 Audit)

Task: DuckDB SQL Query Construction & Parquet Aggregation over Hugging Face dataset (9.8M rows).
Objective: Extract 15-day feature windows (March 1–15) and 16-day outcome windows (March 16–31) without leakage.

2. Six-Step Prompt Iteration Log Summary

Iter #	Technique	Prompt Focus	Observed Output Difference & Impact
1	Naive Baseline	1-line vague request	Fails: missing client grouping, non-functional path, severe target leakage.
2	Role Assignment	Senior Data Engineer persona	Corrects path syntax to read_parquet('hf://...') and multi-column grouping.
3	Context & Motivation	Leakage guard & DATEDIFF rules	Constructs CTE isolating March 1-15 vs March 16-31; joins dim_content metadata.
4	Few-Shot Examples	Input/Output SQL schema pairs	Adds COALESCE(pos, 15.0), division-by-zero protection (+1e-5), and HAVING filter.
5	Output Structure	SQL code block + DuckDB secret	Outputs CREATE SECRET hf syntax automatically; strips conversational prose.
6	Step Decomposition	Chain-of-thought 5-step checklist	Eliminates edge-case mistakes; adds rounded CTR and explicit verification plan.

3. Final Engineered DuckDB SQL Query (Iteration 6)

```
-- Step 1: DuckDB Secret Setup for Hugging Face CREATE SECRET hf (TYPE HUGGINGFACE, TOKEN '${HF_TOKEN}'); -- Step 2-5:
Leakage-Free Temporal Feature Extraction WITH features_raw AS ( SELECT client_hash_id, content_hash_id, SUM(CASE WHEN
report_date <= '2026-03-15' THEN gsc_impressions ELSE 0 END) AS imp_feat, SUM(CASE WHEN report_date <= '2026-03-15'
THEN gsc_clicks ELSE 0 END) AS clk_feat, AVG(CASE WHEN report_date <= '2026-03-15' THEN gsc_avg_position END) AS
pos_feat, SUM(CASE WHEN report_date > '2026-03-15' THEN gsc_impressions ELSE 0 END) AS imp_out FROM
read_parquet('hf://datasets/FlyRank/internship-warehouse/fact_content_daily_performance/month=2026-03/*.parquet')
GROUP BY 1, 2 HAVING SUM(CASE WHEN report_date <= '2026-03-15' THEN gsc_impressions ELSE 0 END) >= 10 ) SELECT
f.client_hash_id, f.content_hash_id, f.imp_feat, f.clk_feat, COALESCE(f.pos_feat, 15.0) AS pos_feat, ROUND(f.clk_feat
/ (f.imp_feat + 1e-5), 5) AS ctr_feat, f.imp_out, DATEDIFF('day', c.content_created_date, DATE '2026-03-15') AS
content_age_days FROM features_raw f JOIN
read_parquet('hf://datasets/FlyRank/internship-warehouse/dim_content.parquet') c ON f.content_hash_id =
c.content_hash_id;
```

4. Cross-Model Audit (Claude 3.5/3.7 vs ChatGPT GPT-4o)

Dimension	Claude 3.5 Sonnet / 3.7	ChatGPT (GPT-4o)
DuckDB Secret Syntax	Exact: Native CREATE SECRET hf used.	Used older SET hf_token='...' syntax.
Leakage Guards	Separated feature vs outcome window inline.	Correct CASE WHEN, but comments placed outside.
Edge-Case Handling	Added COALESCE & continuous 1e-5 epsilon.	Used NULLIF(imp, 0) ratio protection.
Execution Readiness	Ready to paste directly into Python.	Required updating secret format to DuckDB 1.0+.

5. Reusable Prompt Template

Role: Principal Data Engineer specializing in DuckDB & Leakage-Free ML Feature Engineering.
Context: Extract features from [DATASET_PATH] for split cutoff date [CUTOFF_DATE].

Steps required:

- Define authentication / secret for remote Parquet storage.
- Aggregate feature window metrics (report_date <= CUTOFF_DATE).
- Aggregate target window metrics (report_date > CUTOFF_DATE) into label column.
- Join metadata table to compute age via DATEDIFF relative to CUTOFF_DATE.
- Apply COALESCE for null positions and 1e-5 epsilon for ratio features.

Output: Brief 'Decomposition Plan' followed by executable ```sql code block.